SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

FOOD DIGITAL MENU (FDM) SYSTEM

A functional and technical requirements document for a digital restaurant and hotel food-ordering system.

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1. System Overview

The **Food Digital Menu (FDM)** is a digital food ordering and management system designed for restaurants and hotels. The system enables waiters to manage customer food orders using tablets or smartphones.

Customers are not required to download or install any application. Instead, waiters present the digital menu to customers using their own devices, collect individual orders under each customer's name, and submit the orders directly to the kitchen.

The system is intended to streamline restaurant operations, improve order accuracy, facilitate real-time communication between waiters and kitchen staff, and maintain accurate sales records.

2. System Users and Permissions (User Roles & Permissions)

2.1 Waiter

The waiter shall have permission to:

- Display the digital menu to customers using the waiter's device.
- Create orders by selecting a table number and entering customer names.
- Select multiple food and beverage items for each customer.
- Confirm or cancel orders based on customer requests, subject to the cancellation rules defined in this specification.
- Submit completed orders directly to the kitchen.
- View order statuses and updates.

2.2 Kitchen Staff (Kitchen Display System – KDS)

Kitchen staff shall have permission to:

- View incoming orders in real time.
- Identify orders by table number and individual customer names.
- View ordered food and beverage items and their descriptions.
- Receive notifications when new orders are submitted.
- Receive notifications when an order item is cancelled.
- Track incoming orders according to their preparation status.

2.3 Administrator / Restaurant Owner (Admin)

The administrator shall have permission to:

- Access complete sales history and unlimited historical sales records.
- View sales reports, financial summaries, and system statistics.
- Manage, add, edit, and remove menu items.
- Update food and beverage prices.
- Manage menu images and item descriptions.
- Access sales records across all tables, waiters, customers, and time periods.

2.4 Data Protection and Record Security

Only the Administrator shall have permission to delete or modify recorded sales transactions. Waiters and ordinary staff members shall not be allowed to alter or delete finalized sales records. All administrative changes to sales records must be traceable through an audit log.

3. Core Functional Requirements and System Workflow

3.1 Digital Menu Interface and Food Selection

Food Images and Pricing

- Display food and beverage items with high-quality images.
- Display the corresponding price for each menu item.
- Organize menu items into categories for easier navigation.
- Display item descriptions where applicable.

Multiple Food Selection and Automatic Marking

- The waiter shall be able to select multiple food and beverage items for each customer.
- Whenever an item is selected, the system shall automatically mark it with a selection indicator (tick/checkmark).
- The system shall automatically calculate the total cost of all selected items in real time.

Example:

Food Item	Price (TSH)	Selection
Pizza	19,000	Selected
Chicken	10,000	Selected
Total	**29,000**	

3.2 Table and Multiple Customer Management

One Table – Multiple Customers

The system shall support multiple customers placing individual orders under the same table number. The workflow shall be as follows:

- The waiter selects a table number (e.g., Table No. 1).
- The waiter enters the name of each customer.
- The waiter selects food and beverages for each individual customer.
- The system automatically calculates each customer's individual order total.
- The system calculates the overall table total.

Example: Table No. 1

Customer	Ordered Items	Individual Total
Fredy	Pizza (19,000) + Chicken (10,000)	TSH 29,000
Musa	Rice (5,000) + Fish (10,000)	TSH 15,000
Anthony	Additional order items	Automatically calculated
Overall Table Total		**Automatically calculated**

Each customer's order must remain separately identifiable, even when multiple customers share the same table.

3.3 Order Submission and Kitchen Integration (Real-Time Order Sharing)

Once the waiter has completed entering the orders for all customers at a table, the waiter shall click the **“Send to Kitchen”** button.

The system shall immediately transmit the order details to the Kitchen Display System (KDS) or the kitchen's designated order management device.

The kitchen shall receive a clearly organized order summary containing:

- Table number.
- Individual customer names.
- Food and beverage items ordered by each customer.
- Individual customer order totals.
- Overall table order total.
- Order submission time.

The kitchen interface shall update automatically whenever new orders or order modifications are received.

3.4 Order Cancellation Management

The system shall support order item cancellation under the following conditions:

- A customer may request cancellation before the food item has entered preparation.
- The waiter shall be able to cancel the specific item in the system.
- The system shall automatically recalculate the customer's order total.
- The system shall automatically recalculate the overall table total.
- The kitchen shall receive an immediate cancellation notification identifying the affected item, customer, and table number.

Cancelled items must remain identifiable in the order history for accountability and auditing purposes.

The system shall prevent unauthorized cancellation of items that have already entered preparation, unless an authorized administrator or kitchen manager approves an exception.

3.5 Order History and Financial Reporting

Automatic Order History Storage

All completed and finalized orders shall automatically be stored in the system database. The stored records shall include:

- Order identification number.
- Table number.
- Customer names.
- Ordered food and beverage items.
- Item quantities and prices.
- Individual customer totals.
- Overall table totals.
- Waiter identification.
- Order date and time.
- Cancellation records, where applicable.

Data Immutability and Security

- Waiters and ordinary staff members shall not have permission to delete or modify finalized sales records.
- Only authorized Administrators shall have permission to make corrections to recorded sales transactions.

- All administrative changes must be recorded in an audit log, including the administrator's identity, date, time, and reason for the change.
- Original transaction records should remain recoverable for auditing purposes.

General Financial Summary

The system shall provide an administrative dashboard displaying:

- Total recorded sales.
- Total completed orders.
- Sales by table.
- Sales by waiter.
- Sales by individual customer.
- Sales within a selected date or time range.
- Sales breakdown by food and beverage items.
- Cancelled orders and items.

The Administrator shall be able to filter and review sales records using different reporting criteria.

4. Technical Requirements

4.1 Hardware Requirements

The system shall support the following hardware devices:

Waiter Devices

- Smartphones with sufficiently large screens.
- Android tablets or compatible tablets.
- Devices with a stable internet or local network connection.

Kitchen Devices

- Android tablets.
- Dedicated kitchen display screens.
- Small receipt or kitchen order printers, where required.

4.2 Frontend Requirements

The frontend shall be developed as either:

- A Responsive Web Application (recommended), or
- A Native Android Application.

The interface must be optimized for tablet and smartphone screens and shall provide:

- Fast loading and navigation.
- User-friendly menu selection.
- Clear order summaries.
- Automatic price calculations.
- Responsive table and customer management.
- Real-time order status updates.

4.3 Backend and Database Requirements

The backend shall provide a reliable infrastructure for processing, storing, and synchronizing all system data.

The system shall support:

- Centralized database management.
- Real-time order synchronization.
- Secure user authentication.
- Role-based access control.
- Automatic order history storage.
- Secure transaction management.
- Real-time communication between waiter and kitchen interfaces.
- Database backup and recovery mechanisms.

The database must ensure that waiters, kitchen staff, and administrators access consistent and up-to-date information without unnecessary delays.

5. Document Control

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End of Software Requirements Specification (SRS) – Food Digital Menu (FDM) System.